

INDUSTRIAL AUTOMATION ENGINEERING PORTFOLIO

Project 10 — Version 2

Industrial Packaging and Sorting Station

PLC / Simulation Portfolio Project

1. Project Overview

Project 10 Version 2 is the final PLC project in my portfolio and develops the original packaging and sorting application into a structured control system. The station transports packages on a conveyor, counts production, identifies packages that require rejection, operates a reject diverter, verifies the physical reject result, monitors abnormal machine conditions, and stops normal production when a ten-package batch is complete.

I designed the program around separate machine states and command layers rather than allowing field inputs to directly control physical outputs. The final architecture includes Run_Request, System_Ready, Conveyor_Permissive, Auto and Manual mode states, a retained reject cycle, verified reject counting, diagnostic timers, a latched master fault, controlled reset/restart behavior, quality monitoring, and current-batch production statistics.

Manual/Maintenance Mode provides independent conveyor and diverter commands for testing, but the final physical outputs still require Safety_OK and no active master fault. Automatic and Manual control therefore use different command sources while sharing the same common protection layer.

The final PV10 ladder contains Rungs 0000–0042 followed by END at Rung 0043. The project was tested in the PLC/simulation environment and is presented as a controls-engineering portfolio implementation rather than a safety-certified production machine.

2. Engineering Objective

- Separate the momentary Start command from the maintained run request and physical conveyor output.
- Remove Run_Request when Fault_Active becomes true or Safety_OK is lost.
- Require a new Start command after fault or safety recovery.
- Maintain mutually exclusive Automatic and Manual/Maintenance command ownership.
- Use one final physical output rung for the conveyor and one for the reject diverter.
- Count packages only during valid Automatic Mode production.
- Pause the automatic conveyor during an active reject sequence.
- Operate the reject diverter for a defined automatic time interval.
- Require reject confirmation before incrementing the reject counter.
- Supervise reject-cycle duration and reject verification as separate diagnostic conditions.
- Detect package and reject sensors that remain continuously active too long.
- Consolidate machine diagnostics through a latched master fault.
- Treat high reject rate as a quality warning rather than a machine-stopping fault.
- Stop automatic production normally at ten packages.
- Reset batch and recovery states without automatically restarting the machine.
- Maintain total, reject, and good-package production values for future HMI use.

3. Functional Requirements

3.1 Run Control

Start_PB I:1/0 shall latch Run_Request B3:0/0. Stop_PB I:1/1, Fault_Active B3:0/3, or loss of Safety_OK I:1/5 shall remove the run request.

3.2 System Ready

System_Ready B3:0/1 shall require Run_Request, Safety_OK, and no master fault.

3.3 Mode Selection

Auto_Mode_Active B3:3/0 shall require Auto selected and Manual not selected. Manual_Mode_Active B3:3/1 shall require Manual selected and Auto not selected.

3.4 Automatic Conveyor

Conveyor_Permissive B3:0/2 shall require System Ready, Auto Mode, no Batch Complete, and no active reject cycle.

3.5 Package Counting

C5:0 shall count Package_Photoeye I:1/2 transitions only in Auto Mode. The final preset is 10.

3.6 Reject Sequence

Reject_Sensor I:1/3 in Auto Mode shall latch Reject_Cycle_Active B3:1/0. The reject cycle shall pause automatic conveyor operation.

3.7 Reject Diverter Timing

T4:0 shall time the normal automatic diverter interval using Time Base 1.0 and PRE 2.

3.8 Reject Verification

Reject_Confirm_Sensor I:1/6 during an active reject cycle shall latch Reject_Verified B3:1/2. Only Reject_Verified shall increment C5:1.

3.9 Reject Supervision

T4:1 shall supervise total reject-cycle duration using Time Base 1.0 and PRE 5. T4:4 shall supervise receipt of reject confirmation using Time Base 1.0 and PRE 5.

3.10 Sensor Diagnostics

T4:2 and T4:3 shall detect package and reject sensors that remain continuously active for 5 seconds.

3.11 Master Fault

Reject_Timeout, Package_Sensor_Fault, Reject_Sensor_Fault, or Reject_Verification_Fault shall be able to latch Fault_Active B3:0/3.

3.12 Batch Completion

C5:0/DN shall latch Batch_Complete B3:1/1 and activate the batch-complete buzzer and lamp.

3.13 Quality Monitoring

GEQ C5:1.ACC 3 shall establish High_Reject_Rate B3:2/4 and O:2/6 Quality_Alarm_Lamp without feeding the master fault.

3.14 Manual Mode

Manual conveyor and diverter commands shall be hold-to-run and shall remain subject to the common Safety_OK and no-fault conditions at the final output rungs.

3.15 Production Statistics

N7:0 shall mirror C5:0.ACC, N7:1 shall mirror C5:1.ACC, and N7:2 shall calculate N7:0 - N7:1.

4. System Components

Component	Address	Function
Start Push Button	I:1/0	Requests machine operation
Stop Push Button	I:1/1	Removes run request
Package Photoeye	I:1/2	Detects packages for counting
Reject Sensor	I:1/3	Identifies packages requiring rejection
Reset Push Button	I:1/4	Clears applicable fault/batch/reject states
Safety OK	I:1/5	Common safety/permissive status
Reject Confirm Sensor	I:1/6	Confirms physical reject completion
Auto Mode Select	I:1/7	Selects Automatic Mode
Manual Mode Select	I:1/8	Selects Manual/Maintenance Mode
Manual Conveyor PB	I:1/9	Manual conveyor command
Manual Diverter PB	I:1/10	Manual reject-diverter command
Conveyor Motor	O:2/0	Conveyor output
Reject Diverter	O:2/1	Reject actuator output
Batch Complete Buzzer	O:2/2	Audible batch-complete indication
System Running Lamp	O:2/3	System-ready/running indication
Batch Complete Lamp	O:2/4	Visual batch-complete indication
Fault Indicator	O:2/5	Master fault indication
Quality Alarm Lamp	O:2/6	High-reject-rate warning

5. PLC Data and Address Assignment

Address	Symbolic Name	Function / Setting
B3:0/0	Run_Request	Latched operator run request
B3:0/1	System_Ready	Machine ready state
B3:0/2	Conveyor_Permissive	Automatic conveyor permission
B3:0/3	Fault_Active	Latched master fault
B3:1/0	Reject_Cycle_Active	Active automatic reject cycle
B3:1/1	Batch_Complete	Ten-package batch complete
B3:1/2	Reject_Verified	Stored reject confirmation
B3:2/0	Reject_Timeout	Reject cycle exceeded allowed time
B3:2/1	Package_Sensor_Fault	Package photoeye stuck active
B3:2/2	Reject_Sensor_Fault	Reject sensor stuck active
B3:2/3	Reject_Verification_Fault	Reject confirmation missing
B3:2/4	High_Reject_Rate	Quality warning
B3:3/0	Auto_Mode_Active	Valid Auto Mode
B3:3/1	Manual_Mode_Active	Valid Manual Mode
B3:3/2	Manual_Conveyor_Command	Manual conveyor request
B3:3/3	Manual_Diverter_Command	Manual diverter request
T4:0	Reject_Diverter_Timer	TB 1.0, PRE 2
T4:1	Reject_Cycle_Timeout_Timer	TB 1.0, PRE 5
T4:2	Package_Sensor_Stuck_Timer	TB 1.0, PRE 5
T4:3	Reject_Sensor_Stuck_Timer	TB 1.0, PRE 5
T4:4	Reject_Verification_Timer	TB 1.0, PRE 5
C5:0	Total_Package_Count	PRE 10
C5:1	Reject_Package_Count	PRE 32767
N7:0	Total_Packages	Mirrors C5:0.ACC
N7:1	Total_Rejects	Mirrors C5:1.ACC
N7:2	Good_Packages	N7:0 - N7:1

6. Control Strategy

I organized the final control path as Operator / Process Inputs → Operating Mode → Machine State → Automatic or Manual Command → Common Protection → Physical Output → Process Feedback → Verification / Diagnostics.

Start does not directly energize the conveyor. It creates Run_Request, which must pass through Safety_OK and the master-fault condition to establish System_Ready. Auto Mode then establishes Conveyor_Permissive only while the batch is incomplete and no reject cycle is active.

Manual Mode provides an alternate command source for the conveyor and diverter, but the physical output rungs still require Safety_OK and no Fault_Active. Manual control therefore does not bypass the common machine protection layer.

The reject path is Reject Detection → Reject Cycle Active → Conveyor Paused → Diverter Timed → Reject Confirmation → Reject Verified → Reject Count → Reject Cycle Cleared. T4:1 supervises the overall reject cycle, while T4:4 specifically supervises receipt of reject confirmation.

Production information is derived from verified process events. C5:0 counts total packages, C5:1 counts only verified rejects, and N7:2 calculates good packages from those two values.

7. Engineering Design Decisions

Design Decision	Engineering Purpose
Run Request latch	Stores the momentary Start command
Safety loss unlatches Run Request	Prevents automatic restart when Safety_OK returns
Separate System Ready	Distinguishes operator request from current machine authorization
Exclusive Auto/Manual logic	Prevents simultaneous command ownership
Conveyor Permissive	Separates automatic demand from final motor control
Single conveyor output	Combines Auto and Manual commands at one physical coil
Reject Cycle Active latch	Retains sequence ownership after the initiating sensor pulse
Conveyor pause during reject	Coordinates package motion with reject handling
Timed diverter command	Provides a defined automatic actuator interval
Reject confirmation	Verifies the physical process result
Verified reject counter	Prevents unconfirmed rejects from entering production statistics
Separate cycle and verification timers	Distinguishes two different reject failure modes
Sensor stuck diagnostics	Detects abnormal continuously active field inputs
Master fault latch	Provides a common shutdown response while retaining specific causes
Reset separate from Start	Makes recovery deliberate
Quality warning separate from fault	Allows production to continue during a quality concern
N7 production registers	Provides straightforward values for future HMI use

8. Ladder Logic Description

8.1 Rung 0000 — Start / Latch Run Request

Ladder Logic

```
|-----[ XIC I:1/0 ]----- ( OTL B3:0/0 )-----|
      Start_PB                      Run_Request
```

Rung Description:

Rung 0000 latches Run_Request B3:0/0 when Start_PB I:1/0 is pressed. Since the output uses an OTL instruction, the bit stays ON after the momentary Start pushbutton is released.

Engineering Purpose:

I keep the operator's Start command as an internal run request instead of using the pushbutton to control the conveyor directly. This gives the PLC a maintained operating request that can later be removed by Stop, a fault, or loss of the safety permissive.

8.2 Rung 0001 — Stop / Unlatch Run Request

Ladder Logic

```
|-----[ XIC I:1/1 ]----- ( OTU B3:0/0 )-----|
      Stop_PB                      Run_Request
```

Rung Description:

Pressing Stop_PB I:1/1 unlatches Run_Request B3:0/0.

Engineering Purpose:

Stop removes the stored request for machine operation. The downstream ready and conveyor logic then drops out through the normal control path.

8.3 Rung 0002 — Run Request Shutdown - Fault or Safety Loss

Ladder Logic

```
      +----[ XIC B3:0/3 ]-----+
      |      Fault_Active      |
|-----+                      +---- ( OTU B3:0/0 )-----|
      |                      |      Run_Request
      +----[ XIO I:1/5 ]-----+
      Safety_OK
```

Rung Description:

Rung 0002 removes Run_Request B3:0/0 if Fault_Active B3:0/3 becomes true or if Safety_OK I:1/5 is lost.

Engineering Purpose:

I wanted both a master fault and loss of the common safety permissive to cancel the stored run request. When either condition later returns to normal, the machine remains stopped until the operator deliberately presses Start again.

8.4 Rung 0003 — System Ready

Ladder Logic

```
|-----[ XIC B3:0/0 ]-----[ XIC I:1/5 ]-----[ XIO B3:0/3 ]----- ( OTE B3:0/1 )-----|
      Run_Request      Safety_OK      Fault_Active                      System_Ready
```

Rung Description:

System_Ready B3:0/1 is true only while Run Request is active, Safety_OK is present, and the master fault is OFF.

Engineering Purpose:

Run Request represents what the operator wants; System Ready represents whether the controller currently allows operation. Keeping those two states separate makes the startup and shutdown behavior clearer.

8.5 Rung 0004 — Auto Mode Active

Ladder Logic

```
|-----[ XIC I:1/7 ]-----[ XIO I:1/8 ]----- ( OTE B3:3/0 )-----|
      Auto_Mode_Select      Manual_Mode_Select      Auto_Mode_Active
```

Rung Description:

Auto_Mode_Active B3:3/0 turns ON only when the Auto selector is ON and the Manual selector is OFF.

Engineering Purpose:

The automatic production sequence should have command ownership only when Auto is selected by itself. If both selectors are ON, this rung stays false rather than allowing ambiguous mode control.

8.6 Rung 0005 — Manual Mode Active

Ladder Logic

```
|-----[ XIC I:1/8 ]-----[ XIO I:1/7 ]----- ( OTE B3:3/1 )-----|
      Manual_Mode_Select      Auto_Mode_Select      Manual_Mode_Active
```

Rung Description:

Manual_Mode_Active B3:3/1 is true only when Manual is selected and Auto is not selected.

Engineering Purpose:

This is the matching command-ownership logic for maintenance operation. The two mode-active bits cannot intentionally be ON together.

8.7 Rung 0006 — Automatic Conveyor Permissive

Ladder Logic

```
|--[ XIC B3:0/1 ]----[ XIC B3:3/0 ]----[ XIO B3:1/1 ]----[ XIO B3:1/0 ]----- ( OTE B3:0/2 )--|
      System Ready      Auto Mode Active      Batch Complete      Reject Cycle Active      Conveyor Permissive
```

Rung Description:

Rung 0006 establishes Conveyor_Permissive B3:0/2 during normal automatic production. System Ready and Auto Mode must be active, while Batch Complete and Reject Cycle Active must both be OFF.

Engineering Purpose:

The conveyor needs to pause automatically during reject handling and stop normally when the batch is complete. I handle those conditions at the automatic-permissive level instead of forcing them directly into several physical output rungs.

8.8 Rung 0007 — Final Conveyor Motor Control

Ladder Logic

```
      +-----[ XIC B3:0/2 ]-----+
      |      Conveyor_Permissive      |
|-----+                               +-----[ XIC I:1/5 ]----[ XIO B3:0/3 ]---- ( OTE O:2/0 )----|
      |                               |      Safety_OK      Fault_Active      Conveyor_Motor
      +-----[ XIC B3:3/2 ]-----+
      |      Manual_Conveyor_Command      |
```

Rung Description:

Conveyor_Motor O:2/0 can be requested either by the automatic Conveyor Permissive or by Manual_Conveyor_Command B3:3/2. Whichever command path is used, Safety_OK I:1/5 must remain true and Fault_Active B3:0/3 must remain false.

Engineering Purpose:

I kept one physical conveyor output for both operating modes. Manual control therefore changes the command source, but it does not bypass the common safety and master-fault conditions.

8.9 Rung 0008 — Total Package Counter

Ladder Logic

```
|-----[ XIC B3:3/0 ]-----[ XIC I:1/2 ]-----[ CTU C5:0 ]-----|
      Auto_Mode_Active      Package_Photoeye      Total_Package_Count
                                      Preset: 10
```

Rung Description:

During Auto Mode, each false-to-true transition of Package_Photoeye I:1/2 increments C5:0 Total_Package_Count. The counter preset is 10.

Engineering Purpose:

Production counting belongs to the automatic process, not maintenance operation. Requiring Auto Mode prevents manual conveyor testing from being treated as normal production.

8.10 Rung 0009 — Automatic Reject Detection / Latch Reject Cycle

Ladder Logic

```
|-----[ XIC B3:3/0 ]-----[ XIC I:1/3 ]-----[ OTL B3:1/0 ]-----|
      Auto_Mode_Active      Reject_Sensor      Reject_Cycle_Active
```

Rung Description:

A reject detection in Auto Mode latches Reject_Cycle_Active B3:1/0. Once latched, the reject sequence can continue even after the original Reject Sensor signal turns OFF.

Engineering Purpose:

The reject sensor should start the sequence, not directly control its duration. Storing Reject Cycle Active gives the PLC ownership of the complete reject operation.

8.11 Rung 0010 — Reject Diverter Timer

Ladder Logic

```
|-----[ XIC B3:1/0 ]-----[ TON T4:0 ]-----|
      Reject_Cycle_Active      Reject_Diverter_Timer
                                      Time Base: 1.0
                                      Preset: 2
```

Rung Description:

T4:0 Reject_Diverter_Timer runs while Reject Cycle Active is true. The final ladder uses a 1.0-second time base and a preset of 2 seconds.

Engineering Purpose:

The diverter needs a defined automatic operating interval rather than following the reject sensor directly. T4:0 supplies that controlled actuator timing.

8.12 Rung 0011 — Final Reject Diverter Control

Ladder Logic

```
+-----[ XIC B3:1/0 ]-----[ XIO T4:0/DN ]-----+
|      Reject_Cycle_Active  Timer_Done      |
|-----+                                     +-----[ XIC I:1/5 ]-----[ XIO B3:0/3 ]-----[ OTE O:2/1 ]-----|
|                                     |      Safety_OK      Fault_Active      Reject_Diverter
|-----[ XIC B3:3/3 ]-----+
      Manual_Diverter_Command
```

Rung Description:

In Automatic operation, Reject_Diverter O:2/1 remains energized during Reject Cycle Active until T4:0/DN

becomes true. Manual_Diverter_Command B3:3/3 provides the alternate maintenance command. Safety_OK and no active master fault are required for either path.

Engineering Purpose:

As with the conveyor, I use one final physical output for both Automatic and Manual operation. The common protection contacts apply regardless of which mode is commanding the diverter.

8.13 Rung 0012 — Capture Reject Confirmation

Ladder Logic

```
|-----[ XIC B3:1/0 ]-----[ XIC I:1/6 ]-----[ OTL B3:1/2 ]-----|  
Reject_Cycle_Active   Reject_Confirm_Sensor   Reject_Verified
```

Rung Description:

While a reject cycle is active, Reject_Confirm_Sensor I:1/6 latches Reject_Verified B3:1/2.

Engineering Purpose:

An energized diverter output only proves that the PLC issued a command. I use separate physical confirmation before treating the reject as successfully completed.

8.14 Rung 0013 — Verified Reject Counter

Ladder Logic

```
|-----[ XIC B3:1/2 ]-----[ CTU C5:1 ]-----|  
Reject_Verified   Reject_Package_Count   Preset: 32767
```

Rung Description:

A false-to-true transition of Reject_Verified B3:1/2 increments C5:1 Reject_Package_Count. The final counter preset is 32767.

Engineering Purpose:

Only confirmed rejects are included in the reject statistics. A reject command by itself is not counted as a successful process result.

8.15 Rung 0014 — Complete Reject Cycle After Verification

Ladder Logic

```
|-----[ XIC B3:1/2 ]-----[ OTU B3:1/0 ]-----|  
Reject_Verified   Reject_Cycle_Active
```

Rung Description:

Once Reject Verified is true, this rung unlatches Reject_Cycle_Active B3:1/0.

Engineering Purpose:

Verification marks the successful end of the reject sequence. Clearing the reject-cycle state allows the automatic conveyor permissive to return.

8.16 Rung 0015 — Clear Reject Verified

Ladder Logic

```
|-----[ XIC B3:1/2 ]-----[ OTU B3:1/2 ]-----|  
Reject_Verified   Reject_Verified
```

Rung Description:

Rung 0015 clears Reject_Verified B3:1/2 after it has already been used by the reject counter and reject-cycle completion rungs above.

Engineering Purpose:

Rung order matters here. The verification event must first increment C5:1 and clear the reject cycle before the bit itself is removed.

8.17 Rung 0016 — Batch Complete Latch

Ladder Logic

```
|----[ XIC C5:0/DN ]----- ( OTL B3:1/1 )----|  
Total_Package_Count_Done      Batch_Complete
```

Rung Description:

When C5:0 reaches its preset of 10 packages, the counter Done bit latches Batch_Complete B3:1/1.

Engineering Purpose:

Batch completion is a planned production state rather than a machine fault. Once latched, it stops automatic conveyor production through the Conveyor Permissive rung and remains available for operator indication.

8.18 Rung 0017 — Batch Complete Buzzer

Ladder Logic

```
|----[ XIC B3:1/1 ]----- ( OTE O:2/2 )----|  
Batch_Complete      Batch_Complete_Buzzer
```

Rung Description:

The batch-complete buzzer remains ON while Batch_Complete B3:1/1 is active.

Engineering Purpose:

This gives the operator an audible indication that the ten-package target has been reached.

8.19 Rung 0018 — Batch Complete Lamp

Ladder Logic

```
|----[ XIC B3:1/1 ]----- ( OTE O:2/4 )----|  
Batch_Complete      Batch_Complete_Lamp
```

Rung Description:

Batch_Complete_Lamp O:2/4 follows the Batch Complete state.

Engineering Purpose:

The lamp provides a persistent visual indication of normal batch completion until the operator performs a reset.

8.20 Rung 0019 — System Running Lamp

Ladder Logic

```
|----[ XIC B3:0/1 ]----- ( OTE O:2/3 )----|  
System_Ready      System_Running_Lamp
```

Rung Description:

System_Running_Lamp O:2/3 energizes whenever System_Ready B3:0/1 is true.

Engineering Purpose:

I use this lamp to show that the controller has a valid run request, Safety_OK, and no master fault. It represents machine readiness rather than only the state of the conveyor motor.

8.21 Rung 0020 — Package Photoeye Stuck Timer

Ladder Logic

```
|----[ XIC I:1/2 ]-----[ TON T4:2 ]----|  
Package_Photoeye      Package_Sensor_Stuck_Timer  Time Base: 1.0  Preset: 5 s
```

Rung Description:

If Package_Photoeye I:1/2 remains continuously ON, T4:2 accumulates. The timer reaches Done after 5 seconds.

Engineering Purpose:

A normal package detection should be a relatively short signal. Timing a continuously active photoeye gives the PLC a way to distinguish normal package passage from an abnormal stuck or blocked condition.

8.22 Rung 0021 — Package Sensor Fault

Ladder Logic

```
|----[ XIC T4:2/DN ]----- ( OTE B3:2/1 )----|  
Package_Sensor_Timer_Done      Package_Sensor_Fault
```

Rung Description:

Package_Sensor_Fault B3:2/1 becomes true when the package photoeye supervision timer reaches Done.

Engineering Purpose:

This converts the timing result into a named diagnostic condition that can be included in the master-fault logic.

8.23 Rung 0022 — Reject Sensor Stuck Timer

Ladder Logic

```
|----[ XIC I:1/3 ]-----[ TON T4:3 ]----|  
Reject_Sensor              Reject_Sensor_Stuck_Timer  
                          Time Base: 1.0  
                          Preset: 5
```

Rung Description:

T4:3 accumulates while Reject_Sensor I:1/3 remains continuously ON. The timer preset is 5 seconds.

Engineering Purpose:

The reject sensor initiates the automatic reject sequence, so an abnormal continuously active signal needs to be detected before it can cause unreliable sequence behavior.

8.24 Rung 0023 — Reject Sensor Fault

Ladder Logic

```
|----[ XIC T4:3/DN ]----- ( OTE B3:2/2 )----|  
Reject_Sensor_Timer_Done      Reject_Sensor_Fault
```

Rung Description:

When T4:3 reaches Done, Reject_Sensor_Fault B3:2/2 turns ON.

Engineering Purpose:

This gives the master fault architecture a specific diagnostic for abnormal reject-sensor behavior rather than reporting only a generic machine fault.

8.25 Rung 0024 — Reject Cycle Supervision Timer

Ladder Logic

```
|----[ XIC B3:1/0 ]-----[ TON T4:1 ]----|  
Reject_Cycle_Active          Reject_Cycle_Timeout_Timer  
                          Time Base: 1.0  
                          Preset: 5
```

Rung Description:

T4:1 monitors the total time that Reject_Cycle_Active B3:1/0 remains ON. The final preset is 5 seconds.

Engineering Purpose:

This timer supervises the overall reject sequence, not just the diverter output. A reject cycle that fails to finish within the expected period is treated as an abnormal process condition.

8.26 Rung 0025 — Reject Cycle Timeout Diagnostic

Ladder Logic

```
|-----[ XIC T4:1/DN ]-----|-----[ OTE B3:2/0 ]-----|  
Reject_Cycle_Timeout_Done      Reject_Timeout
```

Rung Description:

Reject_Timeout B3:2/0 becomes true when the overall reject-cycle timer reaches Done.

Engineering Purpose:

This diagnostic identifies a reject sequence that has remained active longer than allowed and provides one of the inputs to the master fault.

8.27 Rung 0026 — Reject Verification Timer

Ladder Logic

```
|-----[ XIC B3:1/0 ]-----[ XIO B3:1/2 ]-----[ TON T4:4 ]-----|  
Reject_Cycle_Active    Reject_Verified      Reject_Verification_Timer  
                                           Time Base: 1.0  
                                           Preset: 5
```

Rung Description:

T4:4 Reject_Verification_Timer runs while a reject cycle is active and Reject_Verified B3:1/2 is still false. The timer preset is 5 seconds.

Engineering Purpose:

This timer answers a different question from T4:1: it specifically checks how long the PLC has waited for physical reject confirmation. If verification arrives, the condition disappears and the TON resets.

8.28 Rung 0027 — Reject Verification Fault

Ladder Logic

```
|-----[ XIC T4:4/DN ]-----|-----[ OTE B3:2/3 ]-----|  
Reject_Verification_Timer_Done      Reject_Verification_Fault
```

Rung Description:

If the reject verification timer reaches Done, Reject_Verification_Fault B3:2/3 becomes true.

Engineering Purpose:

This fault represents a difference between the PLC command and the confirmed physical result. The package was expected to reach the reject path, but the confirmation did not arrive within the allowed time.

8.29 Rung 0028 — Master Fault Latch

Ladder Logic

```
+-----[ XIC B3:2/0 ]-----+  
| Reject_Timeout |  
+-----[ XIC B3:2/1 ]-----+  
| Package_Sensor_Fault |  
|-----[ XIC B3:2/2 ]-----+-----[ OTL B3:0/3 ]-----|  
| Reject_Sensor_Fault | Fault_Active  
+-----[ XIC B3:2/3 ]-----+  
Reject_Verification_Fault
```

Rung Description:

Any of the four implemented machine diagnostics can latch Fault_Active B3:0/3: Reject Timeout, Package Sensor Fault, Reject Sensor Fault, or Reject Verification Fault.

Engineering Purpose:

I keep the individual fault bits so the cause remains identifiable, but use one master fault for the common machine response. Once Fault Active is latched, Rung 0002 removes Run Request and the controlled outputs stop through their normal logic.

8.30 Rung 0029 — Fault Indicator

Ladder Logic

```
|----[ XIC B3:0/3 ]----- ( OTE O:2/5 )----|  
      Fault_Active                Fault_Indicator
```

Rung Description:

Fault_Indicator O:2/5 follows the master Fault Active state.

Engineering Purpose:

The operator gets one common visual indication that the machine is faulted, while the individual B3:2/x diagnostic bits identify the specific cause.

8.31 Rung 0030 — Reset Master Fault

Ladder Logic

```
|----[ XIC I:1/4 ]----- ( OTU B3:0/3 )----|  
      Reset_PB                Fault_Active
```

Rung Description:

Pressing Reset_PB I:1/4 clears the latched master fault.

Engineering Purpose:

Reset acknowledges the stored machine-fault state after the abnormal condition has been corrected. If an underlying OTE diagnostic remains active, the master fault can be generated again.

8.32 Rung 0031 — Reset Run Request

Ladder Logic

```
|----[ XIC I:1/4 ]----- ( OTU B3:0/0 )----|  
      Reset_PB                Run_Request
```

Rung Description:

Reset also unlatches Run_Request B3:0/0.

Engineering Purpose:

I deliberately keep Reset and Start as separate actions. Reset prepares the control system for recovery but leaves the machine stopped until another Start command is issued.

8.33 Rung 0032 — Reset Batch Complete

Ladder Logic

```
|----[ XIC I:1/4 ]----- ( OTU B3:1/1 )----|  
      Reset_PB                Batch Complete
```

Rung Description:

Reset_PB clears the latched Batch Complete state.

Engineering Purpose:

This removes the completed-batch condition before the next production batch begins.

8.34 Rung 0033 — Reset Reject Cycle

Ladder Logic

```
|----[ XIC I:1/4 ]----- ( OTU B3:1/0 )----|  
      Reset_PB                Reject_Cycle_Active
```

Rung Description:

Reset unlatches any remaining Reject_Cycle_Active B3:1/0 state.

Engineering Purpose:

An interrupted or failed reject sequence should not remain stored when the machine is prepared for a new operating cycle.

8.35 Rung 0034 — Reset Total Package Counter**Ladder Logic**

```
|----[ XIC I:1/4 ]-----[ RES C5:0 ]----|
      Reset_PB                Total_Package_Count
```

Rung Description:

Reset executes a RES instruction on C5:0 Total_Package_Count, returning its accumulator and counter status to the initial condition.

Engineering Purpose:

Each production batch begins with a fresh total-package count instead of carrying the previous batch value forward.

8.36 Rung 0035 — Reset Reject Package Counter**Ladder Logic**

```
|----[ XIC I:1/4 ]-----[ RES C5:1 ]----|
      Reset_PB                Reject_Package_Count
```

Rung Description:

C5:1 Reject_Package_Count is reset from the same Reset pushbutton.

Engineering Purpose:

The verified-reject count is batch-specific. Clearing it also causes the derived reject statistics and quality-warning condition to return to their initial values.

8.37 Rung 0036 — High Reject Rate Detection**Ladder Logic**

```
|----[ GEQ C5:1.ACC 3 ]-----[ OTE B3:2/4 ]----|
      Reject_Package_Count.ACC >= 3          High_Reject_Rate
```

Rung Description:

The GEQ instruction compares C5:1.ACC with 3. When the verified reject count is three or greater, High_Reject_Rate B3:2/4 turns ON.

Engineering Purpose:

I treat this as a production-quality warning rather than a machine fault. The equipment may be operating correctly even though the batch is producing an unusually high number of rejects.

8.38 Rung 0037 — Quality Alarm Lamp**Ladder Logic**

```
|----[ XIC B3:2/4 ]-----[ OTE O:2/6 ]----|
      High_Reject_Rate          Quality_Alarm_Lamp
```

Rung Description:

Quality_Alarm_Lamp O:2/6 energizes while the High Reject Rate condition is true.

Engineering Purpose:

The operator receives a clear quality warning without forcing a master-fault shutdown. Automatic production can continue until another condition, such as batch completion, stops it.

8.39 Rung 0038 — Manual Conveyor Command

Ladder Logic

```
|-----[ XIC B3:3/1 ]-----[ XIC I:1/9 ]----- ( OTE B3:3/2 )-----|  
Manual_Mode_Active      Manual_Conveyor_PB      Manual_Conveyor_Command
```

Rung Description:

In Manual Mode, holding Manual_Conveyor_PB I:1/9 energizes Manual_Conveyor_Command B3:3/2. Releasing the pushbutton removes the command.

Engineering Purpose:

I keep the maintenance command as an internal hold-to-run request. The physical conveyor output still applies Safety_OK and master-fault protection in Rung 0007.

8.40 Rung 0039 — Manual Diverter Command

Ladder Logic

```
|-----[ XIC B3:3/1 ]-----[ XIC I:1/10 ]----- ( OTE B3:3/3 )-----|  
Manual_Mode_Active      Manual_Diverter_PB      Manual_Diverter_Command
```

Rung Description:

While Manual Mode is active, holding Manual_Diverter_PB I:1/10 energizes Manual_Diverter_Command B3:3/3.

Engineering Purpose:

The diverter can be tested independently of the automatic reject sequence, but its final output remains subject to the common Safety_OK and fault conditions.

8.41 Rung 0040 — Total Package Statistic

Ladder Logic

```
|-----[ MOV ]-----|  
Source: C5:0.ACC  
Dest: N7:0
```

Rung Description:

Rung 0040 continuously copies C5:0.ACC into N7:0.

Engineering Purpose:

I expose the current package total as an integer register so it is straightforward to display later on an HMI or use for higher-level production information.

8.42 Rung 0041 — Total Reject Statistic

Ladder Logic

```
|-----[ MOV ]-----|  
Source: C5:1.ACC  
Dest: N7:1
```

Rung Description:

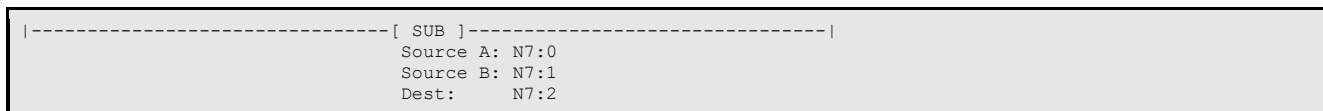
The verified reject counter accumulator is continuously copied into N7:1.

Engineering Purpose:

This gives the production-information layer a simple integer value for the current batch's verified rejects without requiring an HMI to reference the counter structure directly.

8.43 Rung 0042 — Good Package Calculation

Ladder Logic



Rung Description:

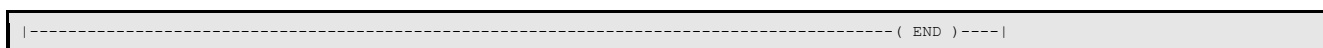
Rung 0042 calculates N7:2 Good_Packages by subtracting N7:1 Total_Rejects from N7:0 Total_Packages.

Engineering Purpose:

The process already knows the total packages and confirmed rejects, so a separate good-package counter is unnecessary. The calculation produces the current good-package quantity directly from the verified production data.

8.44 Rung 0043 — END

Ladder Logic



Rung Description:

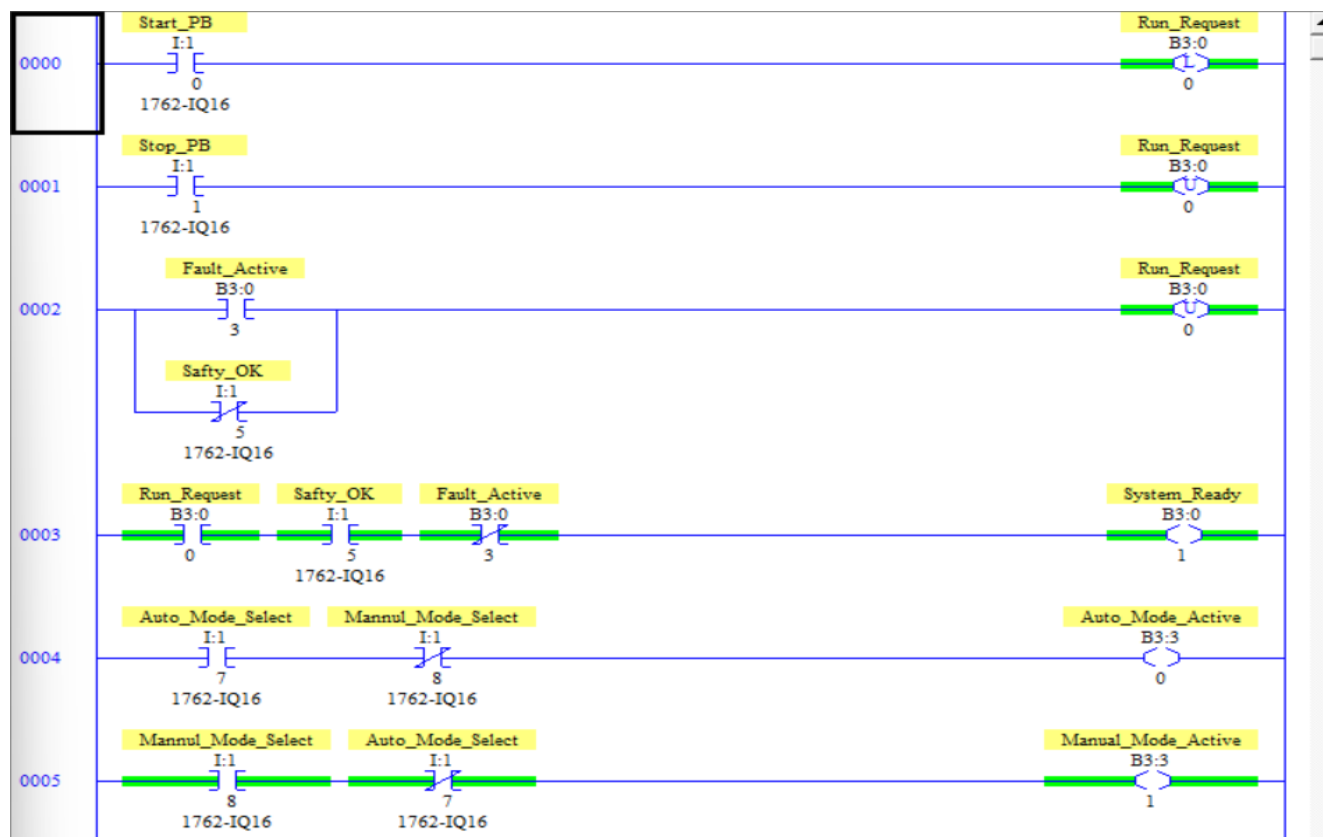
Rung 0043 contains the END instruction and marks the end of the final Project 10 ladder program.

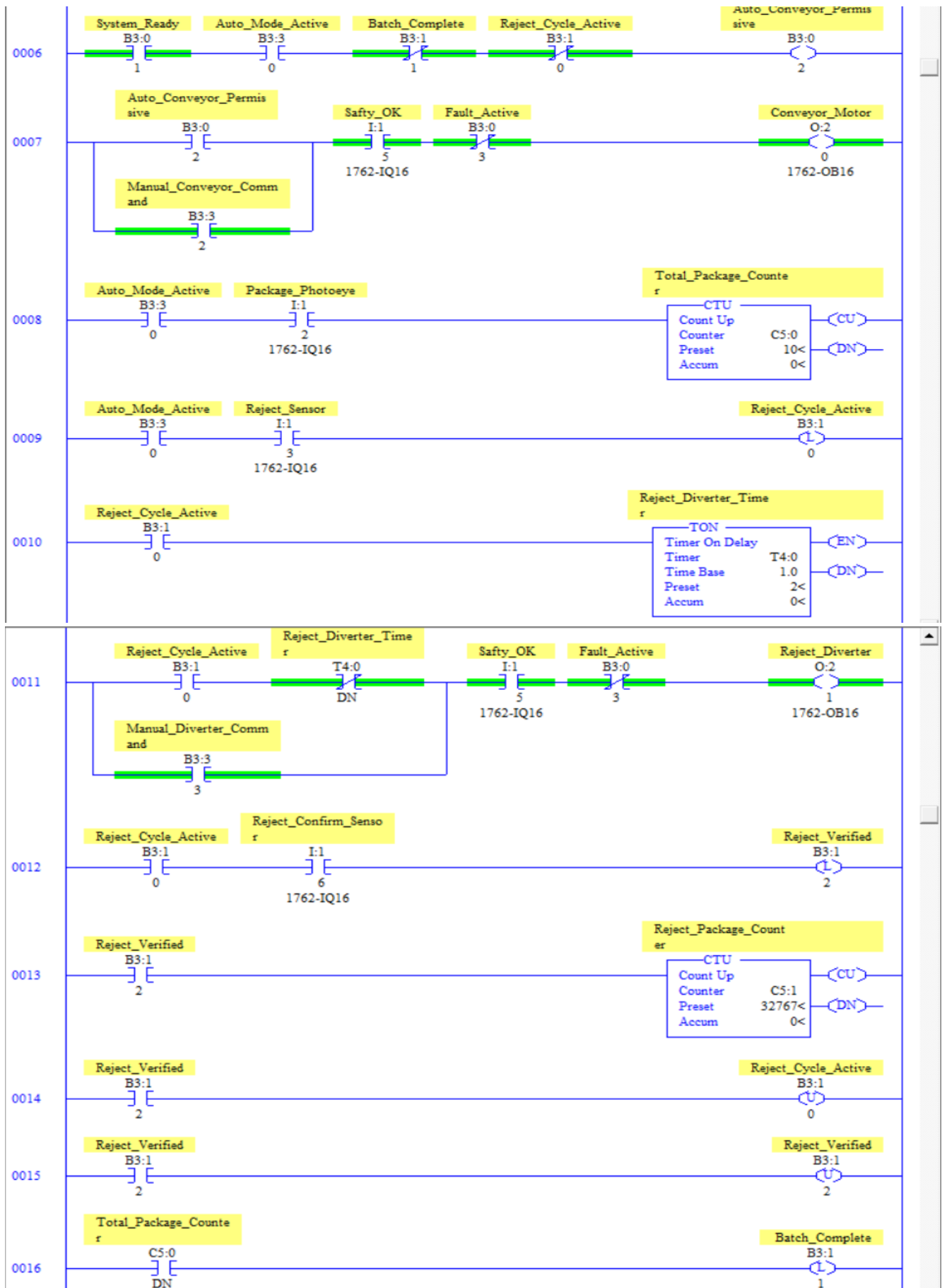
Engineering Purpose:

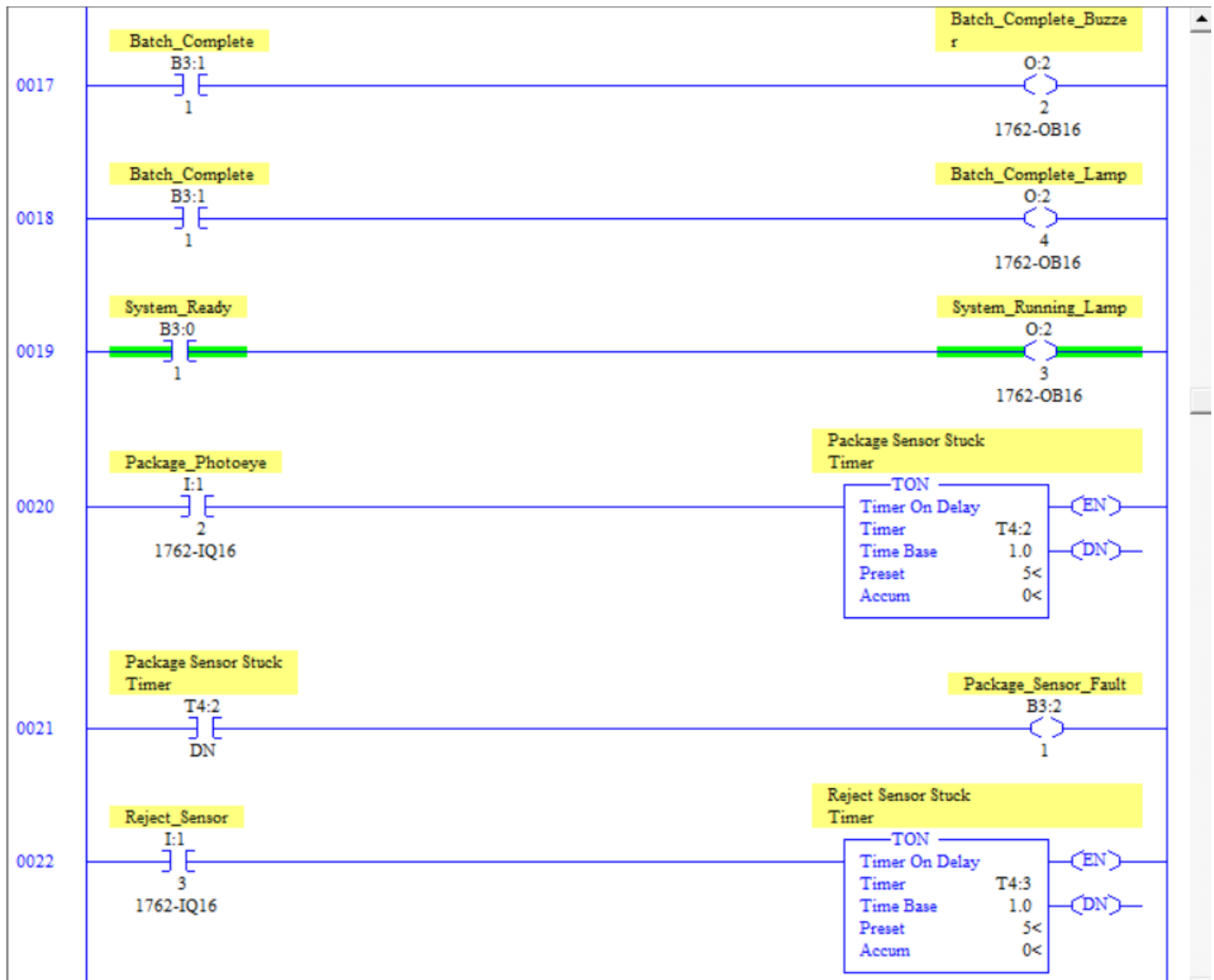
This is the program boundary shown in the final PV10 screenshot. No additional Project 10 ladder logic is implemented below this rung.

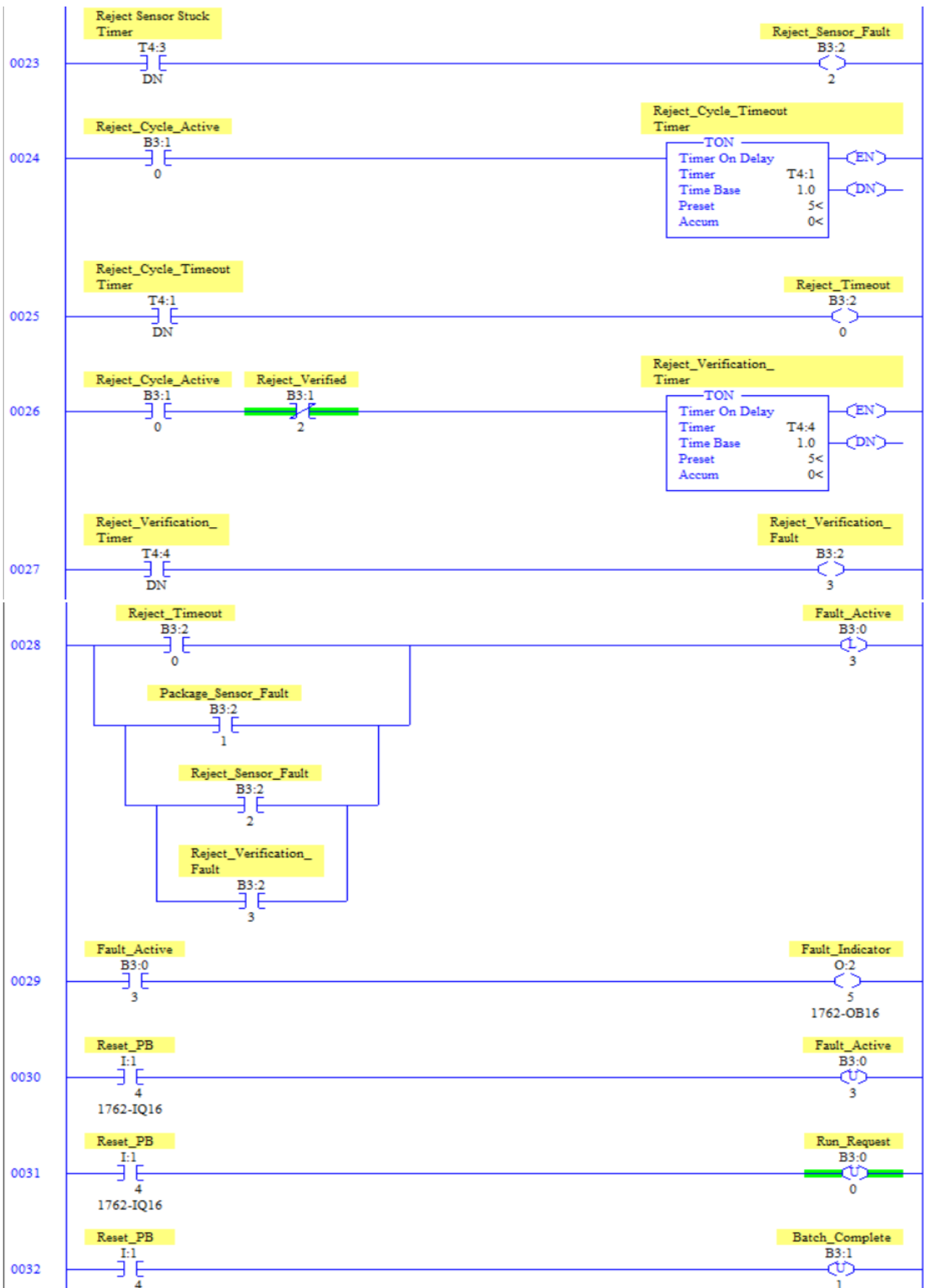
8.45 Actual RSLogix Implementation

The following images are the final Project 10 Version 2 ladder screenshots. They are the source of truth for rung numbering, addresses, branches, timer/counter settings, and implemented logic.









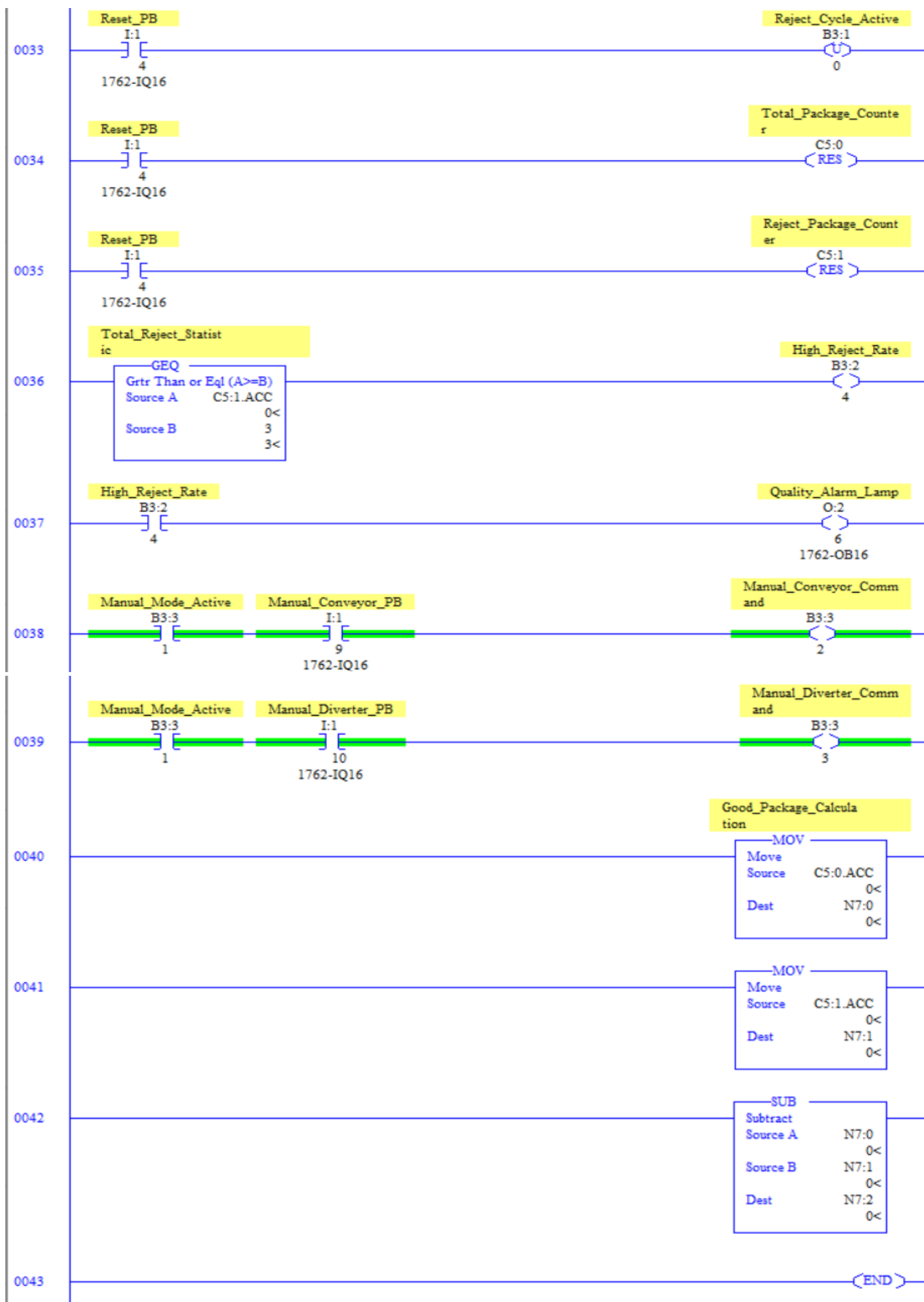


Figure P10V2.1— Final PV10 ladder screenshot.

9. Sequence of Operation

Step 1 — The operator selects a valid operating mode. Auto_Mode_Active becomes true only with Auto selected and Manual not selected; Manual_Mode_Active uses the opposite combination.

Step 2 — Start_PB latches Run_Request B3:0/0.

Step 3 — System_Ready B3:0/1 requires Run Request, Safety_OK, and no master fault.

Step 4 — In Auto Mode, Conveyor_Permissive B3:0/2 becomes true while the batch is incomplete and no reject cycle is active.

Step 5 — The final conveyor rung energizes Conveyor_Motor O:2/0 while Safety_OK remains true and Fault_Active remains false.

Step 6 — Each Package_Photoeye transition in Auto Mode increments C5:0 Total_Package_Count.

Step 7 — A Reject_Sensor event latches Reject_Cycle_Active, which pauses the automatic conveyor.

Step 8 — T4:0 controls the automatic diverter operating interval.

Step 9 — Reject_Confirm_Sensor latches Reject_Verified when the rejected package reaches the reject path.

Step 10 — Reject_Verified increments C5:1, clears Reject_Cycle_Active, and is then cleared by the later rung.

Step 11 — T4:1 supervises total reject-cycle duration while T4:4 supervises receipt of reject confirmation.

Step 12 — T4:2 and T4:3 supervise continuously active package and reject sensors.

Step 13 — Any monitored machine diagnostic can latch Fault_Active, which removes Run_Request through Rung 0002.

Step 14 — When C5:0 reaches 10, Batch_Complete latches and automatic conveyor operation stops normally.

Step 15 — The quality warning becomes active when verified rejects reach three, but production is allowed to continue.

Step 16 — Reset clears the applicable fault, batch, reject, and counter states and removes Run_Request. Start is required for the next operating cycle.

Step 17 — N7:0 mirrors total packages, N7:1 mirrors verified rejects, and N7:2 calculates good packages.

10. Testing and Validation

I tested Project 10 progressively rather than attempting to validate the complete application in one step. The final ladder was frozen only after the required functional and integrated tests passed.

Test	Verification	Result
Initial Reset	Machine states and counters returned to the initial condition	PASS
Auto/Manual Selection	Mutually exclusive mode behavior	PASS
Automatic Startup	Run Request → Ready → Conveyor	PASS
Package Counting	C5:0 incremented correctly	PASS
Successful Reject	Reject cycle, diverter, confirmation, recovery	PASS
Verified Reject Count	C5:1 incremented only after confirmation	PASS
Reject Verification Failure	Missing confirmation generated fault	PASS
Reject Cycle Timeout	Excessive cycle duration generated fault	PASS
Package Sensor Stuck	Package sensor diagnostic generated	PASS
Reject Sensor Stuck	Reject sensor diagnostic generated	PASS

Master Fault	Individual diagnostics produced common shutdown	PASS
Fault Reset	Fault/reject states recovered correctly	PASS
Safety Loss	Run Request removed and outputs stopped	PASS
Safety Recovery	Conveyor did not automatically restart	PASS
Manual Conveyor	Hold-to-run manual conveyor operation	PASS
Manual Diverter	Hold-to-run manual diverter operation	PASS
Manual Protection	Safety/fault conditions still inhibited outputs	PASS
Manual Data Integrity	Manual operation did not alter production counts	PASS
Quality Warning	Three verified rejects activated alarm without fault	PASS
Production Statistics	Total, reject, and good values correct	PASS
Batch Completion	Ten packages stopped automatic production	PASS
Batch Reset	Counters and batch states returned to initial values	PASS
Integrated Production	Complete automatic sequence operated correctly	PASS
Integrated Fault Recovery	Fault → Reset → separate Start recovery	PASS

The final integrated batch produced 10 Total Packages, 3 Verified Rejects, and 7 Good Packages. At that condition, Batch Complete and the quality warning were active while Fault_Active remained OFF.

Validation was performed in the PLC/simulation environment using simulated field conditions. These results verify the implemented ladder behavior; they are not physical machine commissioning or personnel-safety certification.

11. Results

Project 10 Version 2 successfully combines normal machine control, reject verification, diagnostics, Manual Mode, quality monitoring, batch production, controlled recovery, and production information within one PLC application.

A major result is the separation between a PLC command and a verified physical process result. The reject path is Reject Detection → Reject Cycle → Diverter Command → Reject Confirmation → Verified Reject. Only the verified result increments the reject counter.

The application also represents three different outcomes with three different responses: machine faults stop production, a high reject rate generates a quality warning while production continues, and Batch Complete produces a planned normal stop.

The current-batch production layer also operated as intended: N7:0 represents Total Packages, N7:1 represents Total Rejects, and N7:2 represents Good Packages.

12. Challenges and Troubleshooting

Symptom	Diagnostic Path
Machine will not start	B3:0/0 → I:1/5 → B3:0/3 → B3:0/1
Auto conveyor not running	B3:3/0 → B3:1/1 → B3:1/0 → B3:0/2 → O:2/0
Conveyor stopped unexpectedly	Safety_OK → Fault_Active → Batch_Complete → Reject_Cycle_Active
Package count not increasing	B3:3/0 → I:1/2 → C5:0
Reject sequence not starting	B3:3/0 → I:1/3 → B3:1/0
Diverter not operating	B3:1/0 → T4:0/DN → I:1/5 → B3:0/3 → O:2/1
Reject not counted	I:1/6 → B3:1/2 → C5:1
Verification fault	B3:1/0 → B3:1/2 → T4:4 → B3:2/3
Reject timeout	B3:1/0 → T4:1 → B3:2/0
Package sensor fault	I:1/2 → T4:2 → B3:2/1
Reject sensor fault	I:1/3 → T4:3 → B3:2/2
Master fault active	Check B3:2/0 through B3:2/3
Manual conveyor not operating	B3:3/1 → I:1/9 → B3:3/2 → Safety/Fault → O:2/0
Manual diverter not operating	B3:3/1 → I:1/10 → B3:3/3 → Safety/Fault → O:2/1
Quality alarm missing	C5:1.ACC → GEQ 3 → B3:2/4 → O:2/6
Good package count incorrect	C5:0.ACC → N7:0; C5:1.ACC → N7:1; SUB → N7:2

One of the most important commissioning findings involved restart behavior after Safety_OK was lost.

Stopping the output alone was not enough; Run_Request also had to be removed so restoration of Safety_OK could not restart the conveyor automatically.

13. Lessons Learned

This project reinforced that a PLC program has to define more than the normal sequence. I also needed to define command ownership, operating permissives, feedback expectations, diagnostic behavior, machine response to faults, and the operator actions required for recovery.

The reject system was especially useful because it demonstrated the difference between commanding an actuator and confirming the physical result. A successful PLC output does not prove that the package actually reached the reject path.

I also learned the value of separating diagnostic questions. Reject_Timeout asks whether the overall reject cycle remained active too long, while Reject_Verification_Fault asks whether the expected physical confirmation arrived in time.

The quality-warning logic reinforced that not every abnormal production condition should stop the machine. Three verified rejects may indicate a quality issue while the equipment itself is working correctly.

Finally, the Safety_OK test showed why recovery behavior must be tested just as carefully as shutdown behavior. A machine can stop correctly and still restart in an undesirable way if the stored run state is not considered.

14. Industrial Relevance

The architecture used in Project 10 applies to packaging lines, inspection stations, sorting cells, material-handling systems, assembly lines, and other applications that combine production flow with reject handling and machine diagnostics.

- Automatic and Manual command ownership
- stored machine states
- common equipment permissives
- single physical output ownership

- batch production
- sensor supervision
- timed sequence control
- physical process verification
- specific diagnostics and master-fault handling
- controlled recovery and restart prevention
- production statistics and quality-warning logic

A larger industrial implementation could add actuator position feedback, drive/motor status, configurable batch sizes and quality thresholds, HMI alarm messages, alarm history, production orders, defect classifications, SCADA/historian communication, recipes, and safety-rated hardware selected from a formal risk assessment.

15. Version 1 → Version 2 Engineering Progression

Version 1 Foundation	Version 2 Implementation	Engineering Improvement
Basic Start/Stop	Run Request + System Ready	Separates operator request from machine state
Simple conveyor control	Conveyor Permissive	Adds defined automatic operating conditions
Single operating path	Auto/Manual architecture	Separates production and maintenance commands
Direct output logic	Single final output with multiple command sources	Clear output ownership
Basic package count	Ten-package batch control	Defines production batch
Reject sensor → diverter	Retained Reject Cycle	Sequence continues independently of sensor pulse
Timed reject only	Physical reject confirmation	Verifies actual process result
Reject command counted	Verified reject counted	Improves production-data integrity
Limited diagnostics	Four specific diagnostics	Identifies abnormal conditions
No sensor supervision	Package/reject stuck timers	Detects abnormal field signals
Basic reject timing	Diverter + cycle + verification timers	Separates process timing and diagnostics
Generic shutdown	Latched master Fault Active	Common fault response
Output inhibition	Run Request removed on fault/safety loss	Prevents automatic restart
Basic reset	Reset separate from Start	Controlled recovery
No quality monitoring	High Reject Rate warning	Separates quality from machine fault
Limited production information	Total, Reject, Good registers	Supports future HMI/SCADA
Basic simulation testing	Staged + integrated commissioning	Verifies interaction and recovery

16. Project Summary

Project 10 Version 2 — Industrial Packaging and Sorting Station completes the Version 2 PLC project-development phase. The final PV10 ladder contains Rungs 0000–0042 plus END at Rung 0043.

The program includes Run Request, System Ready, safety-loss restart prevention, mutually exclusive Auto/Manual modes, automatic conveyor permission, common Auto/Manual conveyor and diverter outputs, ten-package batch counting, retained reject sequencing, timed diverter operation, physical reject verification, verified reject counting, reject-cycle supervision, reject-verification supervision, package and reject sensor diagnostics, a latched master fault, controlled Reset and separate Start, batch-complete indications, high-reject quality monitoring, and current-batch total/reject/good statistics.

The final integrated commissioning sequence passed normal production, reject handling, reject verification, quality monitoring, batch completion, sensor faults, reject-cycle timeout, verification failure, Manual Mode, Safety_OK loss, fault recovery, and controlled restart.

The most important engineering outcome is the complete control path: Inputs → Machine States → Commands → Permissives → Outputs → Feedback → Verification → Diagnostics → Production Information.

This final project demonstrates my progression from basic ladder programming toward structured industrial controls design in which normal operation, maintenance operation, abnormal conditions, recovery behavior, and production information are considered together.